



Integrating dimensional and scaling analyses with functional modelling and graphs: An approach to comprehend mass transfer in welding

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Abstract

Significant efforts have been made to understand the intricacies of the welding process using numerical methods, machine learning, dimensional, and scaling analyses. Dimensional analysis (DA) is used for qualitative studies of weld pool spreading, heat transfer in welding, welding parameters, and detachment-droplet formation in welding. Nevertheless, DA have been used often in a rather conventional manner. This article proposes to combine DA principles with causally oriented graphical representations and functional analysis to augment the separate capabilities of those methods. The approach uses a physics-based functional model to decompose the welding phenomena into functions, with dimensionless numbers (π) representing aspects of those functions in form of mathematical relationships between the variables. These mathematical relationships are illustrated as a causally oriented graph. This graph is transformed into a system dynamic counterpart. The values of π numbers are estimated using a single example with the developed methodology. The π values in the model are the analogous of biases in Artificial Neural Networks (ANN). The current modelling approach has the advantage of exploiting supplementary sources of knowledge and consequently requires limited data in comparison to supervised machine learning (ML) algorithms used in the field. The proposed methodology is demonstrated with a case study of gas metal arc welding (GMAW) for mild steel. The developed model predicts the droplet formation in GMAW with high accuracy and offer multiple possibilities for extension and generalization to other welding and additive manufacturing processes.

Keywords Gas metal arc welding (GMAW) · Dimensional analysis · Functional modelling · System dynamics · Applied artificial intelligence

1 Introduction

The welding process involves a complex array of physical phenomena, forces, and materials. Alvarez et al. [1] shed light on the historical evolution of modelling techniques in the welding processes. Modelling techniques in gas metal arc welding (GMAW) involve numerical methods, machine learning algorithms, scaling analysis, and dimensional analysis (DA) to simulate weld pool geometry, thermal effects, material properties, welding parameters, arc physics,

residual stress, and weld metal composition [2]. The goal of these models was to optimize welding processes and predict weld characteristics.

The literature surveyed provides an overview of the modelling techniques in welding. Kumar et al. [3] developed a three-dimensional numerical heat-transfer and fluid-flow model to capture the effects of the tilt angle of the fillet joint and the welding positions, on the temperature profiles, velocity fields, weld pool shape, weld pool free surface profile, thermal cycles, and cooling rates during GMAW in spray mode. Zhang et al. [4] studied the temperature and velocity field to calculate the geometry of heat affected zone and weld fusion zone using transient numerical model. Tran et al. [5] created a prediction model with deep learning neural network to calculate the relationship between the welding process parameter and the weld bead size. Nele et al. [6] presented a neuro-fuzzy modelling approach to predict

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welding current when wire feed speed, welding gap, and torch speed are provided as input parameters. Mendez et al. [7] proposed an algorithm combining dimensional analysis with statistical regression to obtain scaling laws to represent complex systems. Choi et al. [8] derived dimensionless number to analyse the welding current and droplet detachment size in GMAW. Nguyen et al. [9] presented a study in which dimensional analysis of the GMAW process was performed in order to identify a number of dimensionless groups formulated based on various GMAW process parameters and material properties that could be used to predict when humping or discontinuous weld beads would occur. When they compared the obtained result with experimental data, they found a reasonably good agreement. Lim et al. [10] used numerical simulation to predict the weld cross section and combine this dimensionless number to perform qualitative analysis of weld pool spreading and heat transfer.

The choice of the modelling method depends on the specific goals of analysis, available resources, and the level of detail required. While reviewing the literature, the authors observed that in several articles, DA has been employed along with other modelling techniques. This has been used to understand the influence of the process parameters on weld pool spreading, heat transfer in welding, welding parameters, and detachment-droplet formation [3, 10–16]. DA appears as a prevalent method amongst researchers to perform a qualitative and quantitative studies. Liang et al. [2] stated that welding modelling researchers would benefit from a systematic application of dimensional analysis in the modelling and analysis of welding processes and material processing. Therefore, DA using dimensionless number appears to be the viable choice to create a prediction model for the GMAW.

DA maps the relationship between the variables based on their fundamental dimensions (length, mass, time, temperature, and charge). DA is used to develop dimensionless numbers by combining an appropriate set of variables. Despite innovation proposed in the domain, selecting meaningful dimensionless numbers requires a significant scientific expertise [17, 18] and the Vashy-Buckingham theorem [19] from DA theory does not offer much support in this process. Furthermore, DA theory does not explicitly capture the functional aspect of the system. These factors strongly reduce the scope and impact of DA in design and modelling. Therefore, this article proposes to use an approach that integrates DA with conceptual modelling techniques to supplement the traditional usage of the method. This approach is named dimensional analysis conceptual modelling framework (DACM) [20]. The DACM framework combines functional analysis and causally oriented graph with DA. This provides an insight into the meaningful dimensionless numbers associated with specific functions governing the behaviour of the system.

System dynamics (SD) is a diagrammatic and mathematical approach used for modelling and simulation of complex dynamic phenomena [21]. The diagram format of SD makes it a good candidate for simulation in the context of the article. In this article, the authors developed a combined approach integrating DACM and SD to model the welding process. DACM framework facilitates the identification of the problem variables, the discovery of meaningful relationships between them and the development of causally oriented graphs representing the design objectives of the modelers. Due to its format, a causally oriented graph can be easily transformed in the SD format and can be used to simulate quantitatively the behaviour of complex systems. The relationships between different variables change in the system over time. This modelling and simulation approach offers several advantages and can be seen as a close alternative to Artificial Neural Networks (ANN). Furthermore, due to its ability to cope with small samples size, the model's estimation times are low. It can also support exploration of the design space and could help predicting optimal manufacturing parameters. Ultimately, it can be exploited in control algorithms for manufacturing systems in decision-making, gain insights into the sydimensionless number characterisestem's behaviour, and propose informed decisions resulting in optimized solutions.

The motivation for the work comes from the fact that welding is a multiphysics process involving different phenomena. The relationships between these phenomena make the modelling process challenging, especially for models aiming at generality, transferability to other welding methods, and quantitative predictions [22]. This problem of generality and transferability could be addressed by using conceptual modelling techniques like DACM. The inherent modularity of the DACM approach allows for easy combination and replacement of modelling blocks. This makes the model easily transferrable to other welding or AM technologies and support generalization of the approach. For this reason, DACM can be a suitable method for developing models for the GMAW. The functional model is developed using the DACM framework [23]. The dimensionless numbers developed in manuscript are based on the Vashy-Buckingham theorem [19] and exploit a clustering and selection approach integrated to DACM [20].

The article discusses in detail the modelling and simulation process. The scope of the article is as given below:

- i. Generation of a functional model and listing variables representing the droplet formation in GMAW process.
- ii. Creating dimensionless numbers from the functional representation model and the variables.
- iii. Developing a causally oriented graph derived from the obtained dimensionless numbers.

- iv. Development of a system dynamics model using dimensionless number to quantitatively predict the droplet diameter during the GMAW process.

The manuscript is organized in the following manner, Sect. 2 details the modelling methodology. Section 3 via case study of GMAW exemplifies the modelling and simulation approach. Section 4 interprets the results obtained in the Sect. 3 and discusses its potential applicability in a broader context. Finally, Sect. 5 presents the conclusions and directions for future research.

2 Methodology

In the proposed process modelling approach shown in Fig. 1, the DACM framework is used to generate a causally oriented graph resulting from a functional model and a clustering of key variables and parameters in the form of dimensionless number. In this article dimensionless numbers are represented with Greek letter π . Subsequently, a system dynamics model is derived from these initial representations to simulate and predict parameters of the droplet formation in the mass transfer of the welding process. The details of this approach are as follows.

Step 1: Modelling objective

The objective definition for a model is a concise statement about the purpose, scope, and the target for the model. It aims at consolidating the efforts towards an effective modelling process ensuring focus and alignment with the targets for the model.

Step 2.1: Functional modelling

Functional modelling, a method delineating a system as a sequence of interconnected functions executing operations serves as a pivotal technique. Functional modelling is essential for representing the system's architecture. The modelling combines different functions that work together to complete system's overall objective. An effective

model necessitates the analysis and understanding of the system's behaviour. In the case of the mass transfer in welding, it requires listing and understanding the underlying physics-based laws and formulae. Each function is assigned by an action verb describing their operations. The functional representation used is a time sequence relationship. The functions on the left-hand side occur prior to the functions on the right.

The authors refer to the modelling methodology proposed by Coatanéa [24], and Mokhtarian et al. [25], which utilizes a limited set of functional vocabulary for the modelling process. These normalised sets of functional terms were derived from the work of Hirtz et al. [26] and Karnopp et al. [27]. Furthermore, the authors have referred to the bond graph theory developed by Paynter et al. [28]. Different functional “organs” illustrated in Fig. 2 are used to represent a function. Transformer (TF) converts certain input energy into desired output. The input and the output for the TF block is in the form of flows (f) and efforts (e). Junction blocks (J) are used when there is a need to combine multiple f and e . Source of energy (SE) is used to indicate input source of energy like electrical energy. The GMAW process involves sequence of operations. These operations involve converting the electrical energy into power, to move wire, to melt wire, to detach droplet, to move torch, and to form weld pool. These operations can be effectively mapped as functions represented by the functional “organs” presented above to form a functional representation model. An important aspect is that functional “organs” are themselves associated with elementary laws.

Step 2.2: Listing variables

The functional “organs”, introduced in step 2.1, are associated with the physical laws describing the step-by-step breakdown of the system being modelled. The functions are themselves connected with a set of generic variables categories derived from bond graph theory [27] [28]. These generic variables explain the interaction and energy transfer within the functions. As outlined in Table 1 the

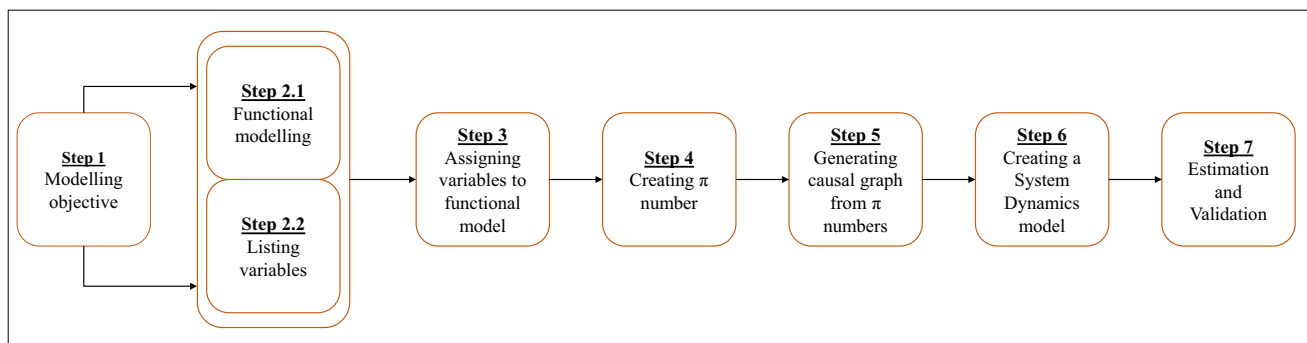


Fig. 1 DACM modelling framework for complex systems

Fig. 2 Functional “organs” in bond graph theory

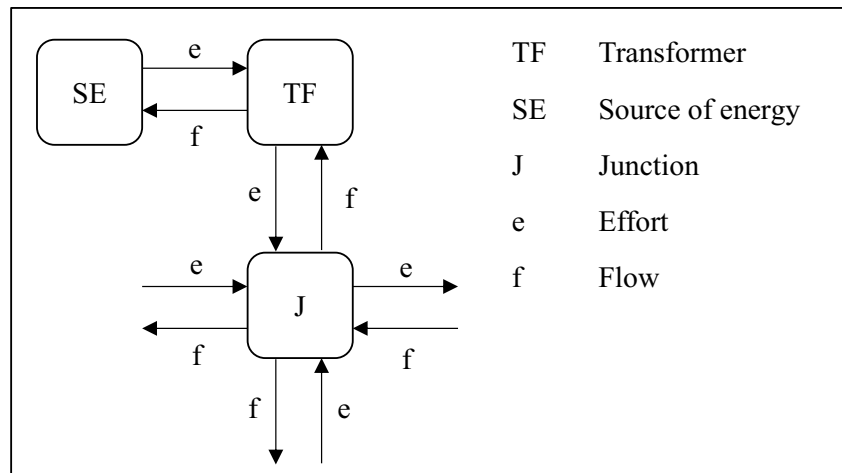


Table 1 Fundamental categories of variables

Primary category of variables	Secondary category of variables
Power variables (P)	Generalized effort (E)
	Generalized flow (F)
State variables (S)	Generalized momentum (M)
	Generalized displacement (D)
	Connecting variables (C)

generic variables are classified into “generalized” flow, “generalized” effort, “generalized” displacement, “generalized” momentum, and connecting variables. The “generalised displacement” is the integration of “flow” over time, whereas “generalised momentum” is the integration of “effort” over time. Furthermore, the functional architecture of the model is incomplete without incorporating the system’s operating environment, materials, geometry, and related aspects. Coatanéa et al. [20] proposed “connecting variable” to encompass other factors not covered in the mentioned categories of the table, such as material and physics properties, geometries, and dimensions. Fig-

ure 3 show the relationship where the state variables are located within the element and the power variables are located outside the element.

Concepts such as energy, efficiency ratio, or acceleration for example can be derived from those elementary generalized variables and for this reason are not represented in the Table 1.

Step 3: Assigning variables to the functional model

Developing a model from a functional representation requires assigning the variables to the functional structure developed at step 2.1. The categorization of the variables proposed in step 2.2 facilitates the assignment of the variables. The power variables are located outside the functional boxes whereas the state variables are located inside the function. Additionally, specific generic environmental variables belonging to the connecting variables category are not assigned to specific functions but remains ubiquitous to the entire system and can then be used in several of the relations described in the following steps. During modelling it is crucial to assign correct variables to specific functions based on the physical phenomenon they represent.

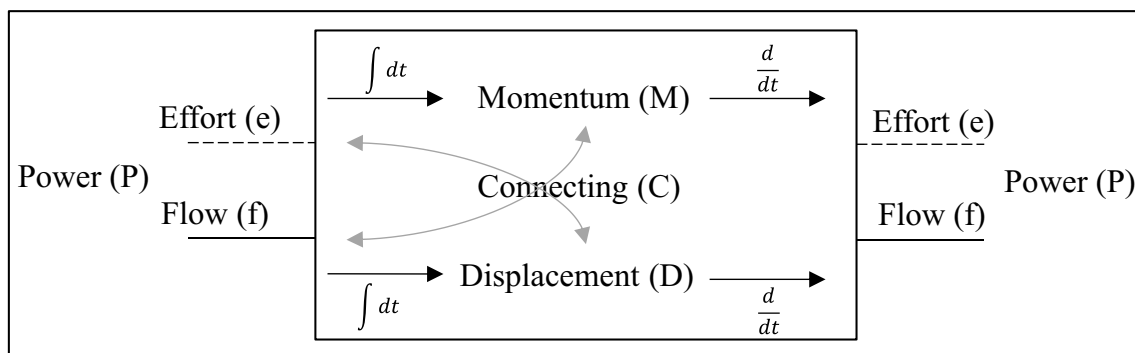


Fig. 3 Representation of the generic variables and their interconnections in the bond graph context

Step 4: Creating π numbers

Dimension analysis (DA) is based on the principle of dimensional homogeneity. The DA approach reduces the complexities in the model to the simplest form by creating relationships between variables based on their fundamental dimensions (mass, time, length, and the four other dimensions recognized in the international system) [29]. Vashy-Buckingham's π -theorem [19] is the widely known theorem formalising a method for computing sets of dimensionless parameters from the given variables. The theory was further developed and presented with application in the field of physics and engineering by prominent researchers [30, 31]. Consequently, this method is adopted for discovering unknown relationships between variables in the present work. Nevertheless, it does not provide any mechanism to discover the useful relationships, and consequently, this is why the previous steps have been added before step 4 for forming meaningful clusters of variables.

A dimensionless number characterise a given physical situation and in this article is associated with one or multiple functions. According to the theorem, every law which takes the form $y_0 = f(x_1, x_2, x_3, \dots, x_n)$ can take the alternative form:

$$\pi_0 = f(\pi_1, \pi_2, \dots, \pi_n) \quad (1)$$

where π is a dimensionless product. This alternative form is the result of the dimensional analysis and is the consequence of the Vashy-Buckingham theorem.

A dimensionless number is a product which takes the following form:

$$\pi_k = y_i \cdot x_j^{\alpha_{ij}} \cdot x_l^{\alpha_{il}} \cdot x_m^{\alpha_{im}} \quad (2)$$

where x_j are repeating variable, y_i is performance variable, and α_{ij} are the coefficient. Equation (2). presents the dimensionless form of reusable, modelling primitives, used to form the dimensionless number π . Examples of those primitives are present in multiple domains of science. For example, the Reynolds number, the Eckert number, and the Weber number are some examples of dimensionless primitives.

Szirtes et al. [32] proposed a methodology to systematically determine the dimensionless number for a set of variables. They divided the repeating and performance variables into matrices represented by alphabets A and B respectively, where B is a non-singular square matrix. Using the formula (3), they determined the coefficients for the repeating variables.

$$C = -(A^{-1} \cdot B)^T \quad (3)$$

Step 5: Generating causally oriented graph from π numbers

Iwasaki and Simon [33] and later Travé-Massuyès [34] developed algorithms to generate causally oriented graphs from a set of equations. Travé-Massuyès algorithm is also able to integrate the different modes of a system. During the same time period, Bhaskar et al. [35] proposed a mathematical machinery based on dimensionless primitives to describe the dynamic behaviour of a system. Implicitly, they also associated dimensionless numbers with the concept of functions. They named a dimensionless number π , referring to a particular function as a regime. They used the repetitions of similar variables within different dimensionless numbers as a propagation mechanism allowing them to describe the dynamic propagations taking place within systems.

The DACM framework integrates those developments and provides additional tools to analyse causally oriented graphs. The variables in those coloured graphs are classified into four categories, depending on their implication on the system. Colours are associated with each category of variables. The variables are as follows.

Exogeneous variables They are imposed onto the system. They are part of the environment of the system. The exogeneous variables are by nature not controllable. Traditionally, the designer cannot modify them, except in some cases where modifying the border of a system might allow to make them controllable. They are presented in grey in causally oriented graph.

Independent variables These variables are not influenced by any other variables in the system. Modellers can modify the value of those design variables with more degrees of freedom. The independent variables are represented in green colour.

Dependent variables These variables, coloured blue, are influenced by other variables and are thus more difficult to control than independent design variables.

Performance variables They are a special class of dependent variables. They are important for the overall performance evaluation of the system from a designer or modeller perspective. The modellers usually define target values or qualitative objectives on those variables. They are coloured in red.

Step 6: Creating a system dynamic model

By step 5, the creation of a causally oriented graph from the functional structure is completed. Concurrently, the DACM framework can be used in a systematic manner to

find the weakness and contradictions within the system. The causal graph provides information about the influence of variables on the performance variable. This information can be used for analysing the model. Mokhtarian et al. [25] used this approach to perform qualitative analysis of the glue gun system. In this article, the authors propose to use the DACM framework combined with system dynamics to perform a quantitative analysis of the process parameters. Two types of models are generated with Vensim [36]. Vensim is a software used to develop and simulate models in the field of system dynamics (SD). The first model will show a cyclic relationship between the performance variable and the π numbers. Any change in the value of π number will result in a change in the value of corresponding performance variable. The objective is to use this model for estimating π values. The second model demonstrates a causal relationship between the performance variables, the associated π numbers, and other variables. In this model, subsequent performance variables are dependent on the prior performance variable, exhibiting causal behaviour. The causally oriented graph forms the blueprint to construct the causal model in Vensim. The second model is then used for prediction. The second model is in essence a trained model that can be used for prediction of system objectives. This concurrent approach enables a quantitative understanding of the variables impact on the performance of the system. Thus, establishing a foundation for quantitative analysis of the system model. The mapping between a DACM causally oriented graph and a SD model is described in the case study of Sect. 3. This transformation can be easily automated to generate SD models.

Step 7: Estimation and validation

In case of the framework presented in this article, the estimation phase aims to estimate the π values that fit the model to the experimental dataset. In other words, the model is tweaked to minimize the difference between the predicted value and the experimental value from the dataset. Thus, the estimated value of the π number is a common analogy with the concept of weight and bias in ANN. The method developed in this article for π value estimation, when compared to supervised learning algorithms requires a substantially reduced amounts of data for training and validating the model. This is a fundamental advantage of the approach presented in this article as it requires minimal data to estimate the π values. Usually in supervised learning applications such as ANN, the training phase involves the use of the well-known back propagation algorithm [37] that fine tunes the model's weights and biases. Unlike ANNs where the weights and biases are adjusted using a back-propagation algorithm, our model relies on the causal ordering and integration of

the conservation principle in our dimensionless numbers to compute the optimal values of the π numbers.

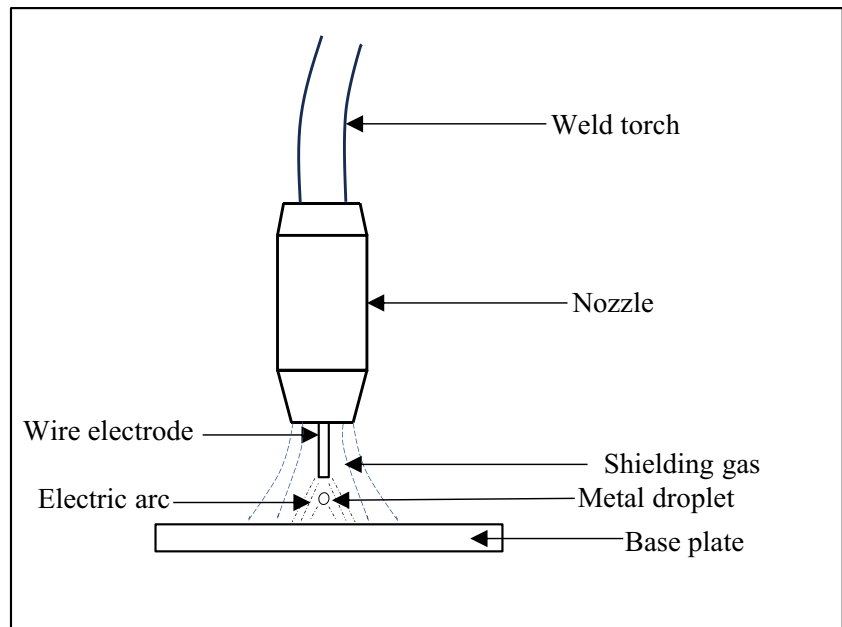
As the values of π numbers and the variables of interest are usually both unknown. The structure of the model is not a self-contained structure i.e. the number of unknowns is greater than the number of equations. In the case of ANNs, the answer to this problem is provided by an increase of the size of the training set. In the context of this article, the data required for estimating the π values is referred as estimation set. A fundamental goal of this framework is to keep the estimation set as small as possible. In the present article, three additional information are brought to maintain the constraint on the size of the estimation dataset. First, the π numbers are built in such a way that the mass transfer studied in this welding case integrates the physics principle of conservation. This then allowing to impose the values of some of the π numbers to be 1. Second, the principle of similarity from DA [31] is used to keep constant those π numbers' values constant after the estimation phase. This implies that the model is able to represent only one behaving mode described by the experimental dataset.

A single example is needed to determine the values of the unknown π numbers such that the predicted value of target variables matches the actual value of the target variable. Thus, using the first model, we can calculate the values of the π numbers. These π values are then entered into the second model. This model exhibits causal behaviour between the variables and the π numbers. It is important to mention that the model used for this purpose is different from the model used for estimating π values, as this model does not contain simultaneous evolving equations. After inputting the π values, the only unknown variables in the causal structure are the performance variables. Therefore, using the model and the input variables, the values of performance variables are calculated, ultimately leading to the target variable. This aspect is described further in the case study.

3 Case study: Prediction of welding droplet size

The case study illustrates the methodology presented in this article to predict the size of droplet during GMAW process. GMAW can be categorised into metal inert gas (MIG)/metal active gas (MAG) welding. In this type of welding process, the filler material is in the form of wire. The GMAW consists of a power source, wire feeder, welding torch, and a shielding gas supply. A schematic presentation of the process is illustrated in Fig. 4. When an electric current passes

Fig. 4 Schematic illustration of the GMAW welding process



through the wire, it encounters electrical resistance, which leads to heating of the wire to reach its ionization energy threshold. This process liberates ions, initiating an electric arc between the wire electrode and the workpiece. Here, the wire electrode acts as anode and the base plate serves as cathode.

The electric arc, as a result of ionization, becomes the locus of intense heat generation, reaching the melting temperature of the wire electrode. As the wire melts, it forms a pool of molten metal. The transfer of the molten metal from the wire electrode to the weld pool is influenced by the electromagnetic force, gravitational force, drag force, and surface tension. This molten metal serves as the filler material that joins the base metals together to create the welded joint. The shielding gas envelops the arc and the molten weld pool, protecting the melt pool from oxidation or contamination. The controlled melting of the wire electrode and the regulation of the welding torch speed and wire feed rate are crucial in determining the size, quality, and characteristics of the resulting weld bead.

Step 1: Modelling objective

Modelling of the welding process begins with defining the system and identifying the scope of modelling. The objective of the model presented in this paper is to predict the droplet size of the molten wire electrode during the GMAW welding process.

Step 2.1: Functional modelling

This step refers to creating a functional presentation of the GMAW. In the functional presentation diagram, the entire process is represented in a sequence of func-

tion satisfying the systems overall operation. The functions involved in droplet formation in GMAW include the following: (a) to move wire, (b) to convert electrical energy into power, (c) to resist current, (d) to melt wire, and (e) to detach droplet. By utilizing the methodology mentioned in step 2.1 of Sect. 2, a functional diagram is constructed. Figure 5 illustrates the functional diagram containing the mapping between the functions involved in the droplet formation in the GMAW process. In the future research, the model will be expanded to predict different dimensions of weld pool.

In the DACM framework, functional “organs” are associated with the linguistic description of the functions. It forms an intermediate step between the linguistic description and the design synthesis. The transformer TF1 converts the source of electrical energy into power, whereas the TF2 represent the movement of the wire in the welding process. Resistor 1 is used to symbolize the resistance in the wire based on its relative position from the base plate and the flow of the shielding gas, when the current flow through the wire ionization process begins which results into an electric arc generating an intense heat. Thus, junction J1 embodies the interaction between TF1, TF2, and R1 symbolizing the heat generation in the GMAW process. Furthermore, the heat generated is becoming sufficient to melt the filler wire as represented in TF 3. After the melting of the wire, there is a phase transformation, and the liquid filler material is deposited on the base material in the form of droplets. The detachment of the drops from the wire electrode is embodied in TF 4.

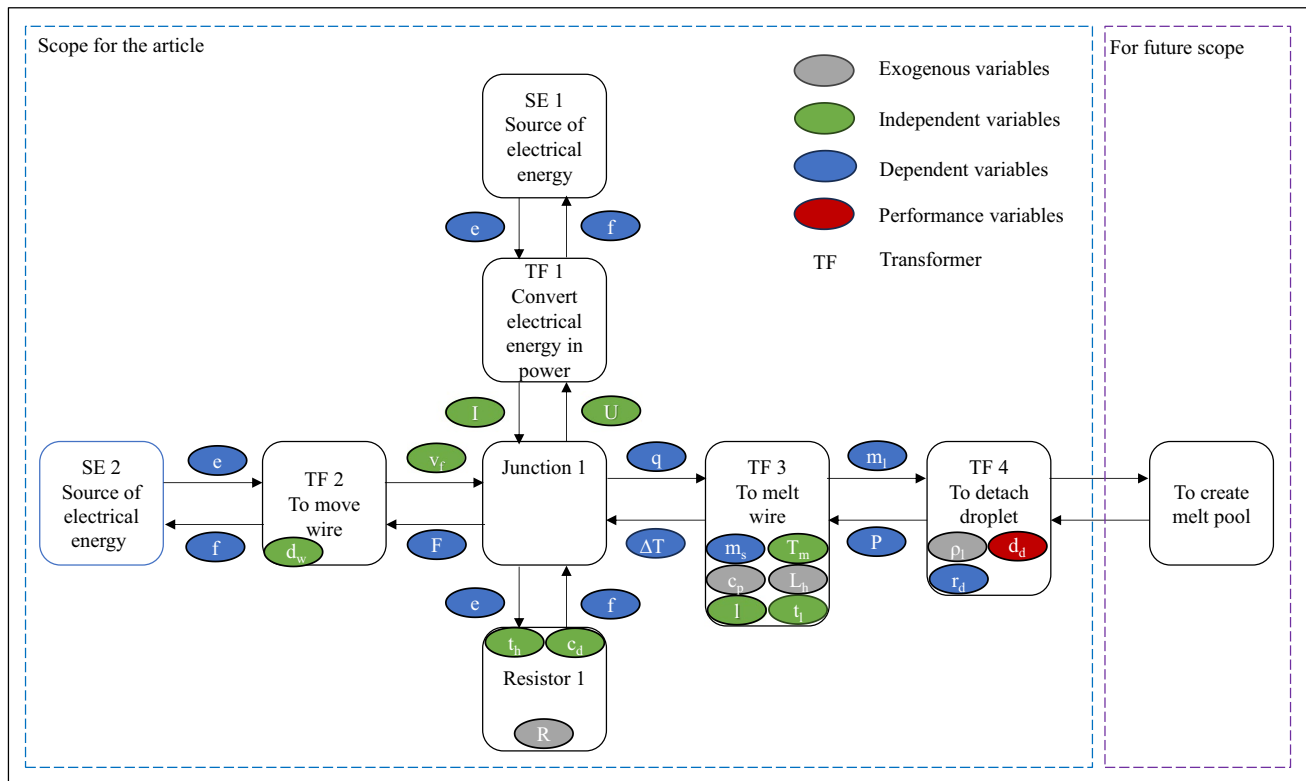


Fig. 5 Functional model for droplet formation in GMAW

Step 2.2 and 3: Listing and assigning variables

The next step in the methodology is to populate the functional representation with variables. The authors have relied on the physical laws and formula encapsulating the GMAW process to assist them in assigning the variables to the corresponding functions [38]. According to step 2.2 Sect. 2, the variables are classified into five classes. Table 2 represents the variables of the model, their associated dimensions, units, and categories. Moreover, as to create a causal relation between the variables. Colours are associated with the variables highlighting their nature and impact on the system. The variables are categorised with the colour code described above in Sect. 2. The variables along with their categorization are presented in the functional representation diagram in Fig. 5. This categorization assists in understanding the impact of the variables on the behaviour of a function.

It should be noted that the model presented in this article does not account for gas flow, magnetic permeability, current polarity, and other variables related to welding. This limitation arises from the unavailability of the dataset containing this information. The authors intend to include these variables in future work as an extension to the current model.

Step 4 and 5: Generating a coloured causally oriented graph using π numbers

The approach described in the book from Szirtes [32] is employed to form the dimensionless numbers associated with the functions from Fig. 5. This allows to define the relevant dimensionless numbers associated with the functions. This is also guiding the choices of the performance variables, as outputs of the functional transformation, when creating the dimensionless numbers.

The generated dimensionless numbers are:

$$\pi_1 = d_w \cdot q_l \cdot I^{-2} \cdot R^{-1} \cdot t_h^{-1} \quad (4)$$

$$\pi_2 = c_d \cdot q_2 \cdot I^{-2} \cdot R^{-1} \cdot t_h^{-1} \quad (5)$$

$$\pi_3 = v_f \cdot q_3 \cdot I^{-2} \cdot R^{-1} \quad (6)$$

$$\pi_4 = m_s^2 \cdot c_p \cdot \Delta T \cdot q^{-2} \quad (7)$$

$$\pi_5 = m_l \cdot L_h \cdot t_l^2 \cdot m_s^{-1} \cdot I^{-2} \quad (8)$$

$$\pi_6 = d_d \cdot \rho_1^{1/3} \cdot r_1^{1/3} \cdot m_l^{-1/3} \quad (9)$$

Table 2 System variables with associated dimensions and categories

Variable	Symbol	Unit	SI base dimensions	Categories
Electric potential	U	Volts	$\text{kgm}^2\text{s}^{-3}\text{A}^{-1}$	Flow
Electric current	I	Amperes	A	Effort
Heat flow rate	q	J/m	$\text{kgm}^1\text{s}^{-2}$	Flow
Temperature difference	$\Delta\mathbf{T}$	K	K	Effort
Wire feed rate	v_f	m/s	ms^{-1}	Flow
Force (move wire)	F	N	kgms^{-1}	Effort
Wire diameter	d_w	m	m	Displacement
Contact tip to work distance (CTWD)	c_d	m	m	Connecting
Time to heat wire	t_h	s	s	Connecting
Resistance	R	Ohms	$\text{kgm}^2\text{s}^{-3}\text{A}^{-2}$	Connecting
Specific heat capacity	C_p	J/kgK	$\text{m}^2\text{s}^{-1}\text{K}^{-1}$	Connecting
Melting temperature	T_m	k	K	Connecting
Mass of solid melted	m_s	kg/s	kg s^{-1}	Connecting
Mass of liquid melted	m_l	kg/s	kg s^{-1}	Connecting
Length of wire melted	l	m	m	Connecting
Time to melt the wire	t_l	s	s	Connecting
Enthalpy of melting	L_h	J/kg	m^2s^{-2}	Connecting
Droplet rate	r_d	1/s	s^{-1}	Flow
Pressure	P	N/m ²	$\text{kgm}^{-1}\text{s}^{-2}$	Effort
Droplet diameter	d_d	M	m	Displacement
Density	ρ_l	kg/m ³	kgm^{-3}	Connecting

The article does not describe in detail the construction of the dimensionless numbers due to the length constraints of the paper. The number of dimensionless numbers formed using the procedure describes in Szirtes' book [32] is constrained by the number of variables and the number of base dimensions describing each cluster. π_1 , π_2 , and π_3 signify the conversion of electrical energy into heat energy. These π numbers are associated with Junction 1, TF1, TF2, and Resistor 1. π_4 represents the ratio between the amount of heat required to melt the material to the supplied heat. This is an implementation of the conservation principle from physics. It is linked via TF3. Similarly, π_5 is also associated with TF3. It represents the phase transition from solid to liquid. π_6 gives information about the droplet formation and is described by TF4. Figure 6 illustrates the relation between the blocks of the functional representation diagram and π numbers. It could be noted that a group of dimensionless numbers can be associated with a group of functions (i.e. π_1 , π_2 , and π_3) or with one single function (i.e. π_4 , π_5 , and π_6). Bhaskar et al. [35] proposed a machinery to analyse the dynamic relationships taking place between dimensionless numbers. This machinery is used later in the paper as the basic mechanism for representing the dynamic connections between dimensionless numbers. The formed dimensionless equations need to be reformulated by taking their outputs on the left side of the equations. The

outputs being the selected performance variables. This is the start of the process leading to causally oriented graphs. Equations (4) to (9) are rewritten to solve for the value of the performance variables.

$$q_1 = \pi_1^1 \cdot d_w^{-1} \cdot I^2 \cdot R^1 \cdot t_h^1 \quad (10)$$

$$q_2 = \pi_2^1 \cdot c_d^{-1} \cdot I^2 \cdot R^1 \cdot t_h^1 \quad (11)$$

$$q_3 = \pi_3^1 \cdot v_f^{-1} \cdot I^2 \cdot R^1 \quad (12)$$

$$m_s = \pi_4^{1/2} \cdot c_p^{-1/2} \cdot \Delta T^{-1/2} \cdot q^1 \quad (13)$$

$$m_l = \pi_5^1 \cdot L_h^{-1} \cdot t_l^{-2} \cdot m_s^1 \cdot l^2 \quad (14)$$

$$d_d = \pi_6^1 \cdot \rho_l^{-1/3} \cdot r_d^{-1/3} \cdot m_l^{1/3} \quad (15)$$

In Eq. (13), q is the summation of the performance variables q_1 , q_2 , and q_3 of Eqs. (10), (11), and (12). Similarly, m_s and m_l are repeating variables in Eqs. (14) and (15) respectively and are performance variables in Eqs. (13) and (14). Those variables are shared between equation and serve as a propagation mechanism between dimensionless numbers. Equations (10) to (15) have shared

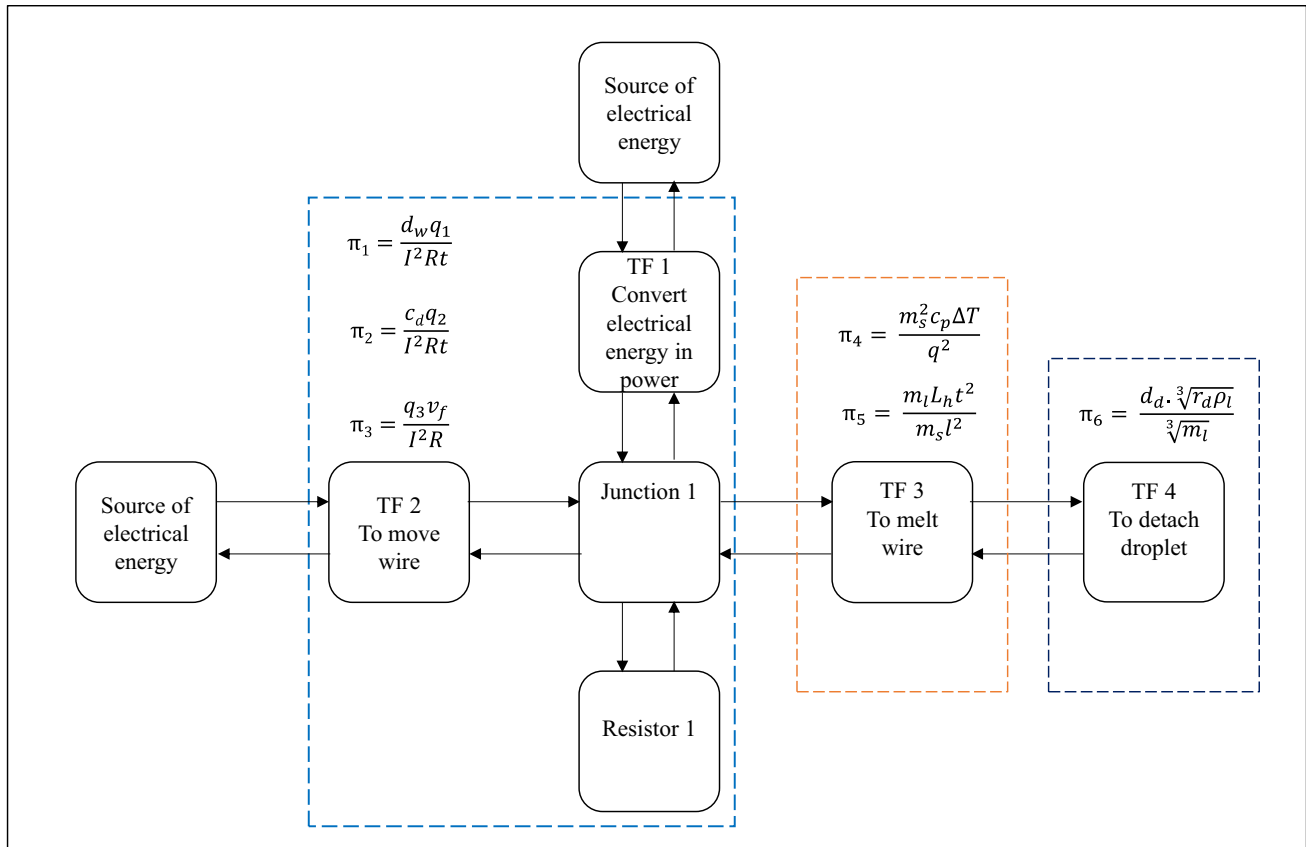


Fig. 6 Functional representation diagram and π numbers relationship

variables useful to propagate the dynamic behaviour of the GMAW process. Using the categorization of variables and the formulations in Eqs. (10) to (15), together with the causal algorithm from Iwasaki and Simon [39], a causally oriented graph can be formed as represented in Fig. 7. It should be noted that a software tool automating the construction of those graphs has been developed by the research group which authored this paper.

The causally oriented graph in Fig. 7 is suitable to predict the droplet diameter if the performance variables are the only unknowns. However, there might be instances where the values of π numbers are also unknown. This prompts for an approach where reducing the number of unknowns can be obtained using supplementary principles and knowledge embedded in the equations. As described above in Sect. 2, the principle of conservation is integrated in the construction of certain dimensionless numbers, and consequently, the related π constants have a value equal to 1. Also, it is assumed that the similarity principle holds true. This means that after estimating the π values in each dimensionless number, those values are considered to remain constant. The estimated values are then valid for a specific functioning mode; change in π

values implies that the system is behaving in a different behavioural mode [34]. The study of the changes in modes is briefly discussed in Sect. 4 of this paper and future developments are proposed.

In addition to the additional principles used to set several π values, the estimation of the other π values necessitates a causal orientation graph with a cyclic relationship between performance variables and respective π numbers. Here the π numbers are expressed as the function of performance variables and other variables, while conversely, the performance variables are expressed as the function of respective π numbers and other variables. For instance, assuming the value of mass of solid (m_s) enables the calculation of π_4 , and similarly, assuming the value of π_4 allows the determination of the mass of solid (m_s). Figure 8 illustrates the section of cyclic causal orientation graph. The readers should note that two causally oriented graphs are produced, Fig. 8 being used for estimation purpose and Fig. 7 used for prediction purpose after the estimation phase. Figure 8 when implemented in SD format on the Vensim software is necessitating the use of simultaneous equations. This is presented in the following step.

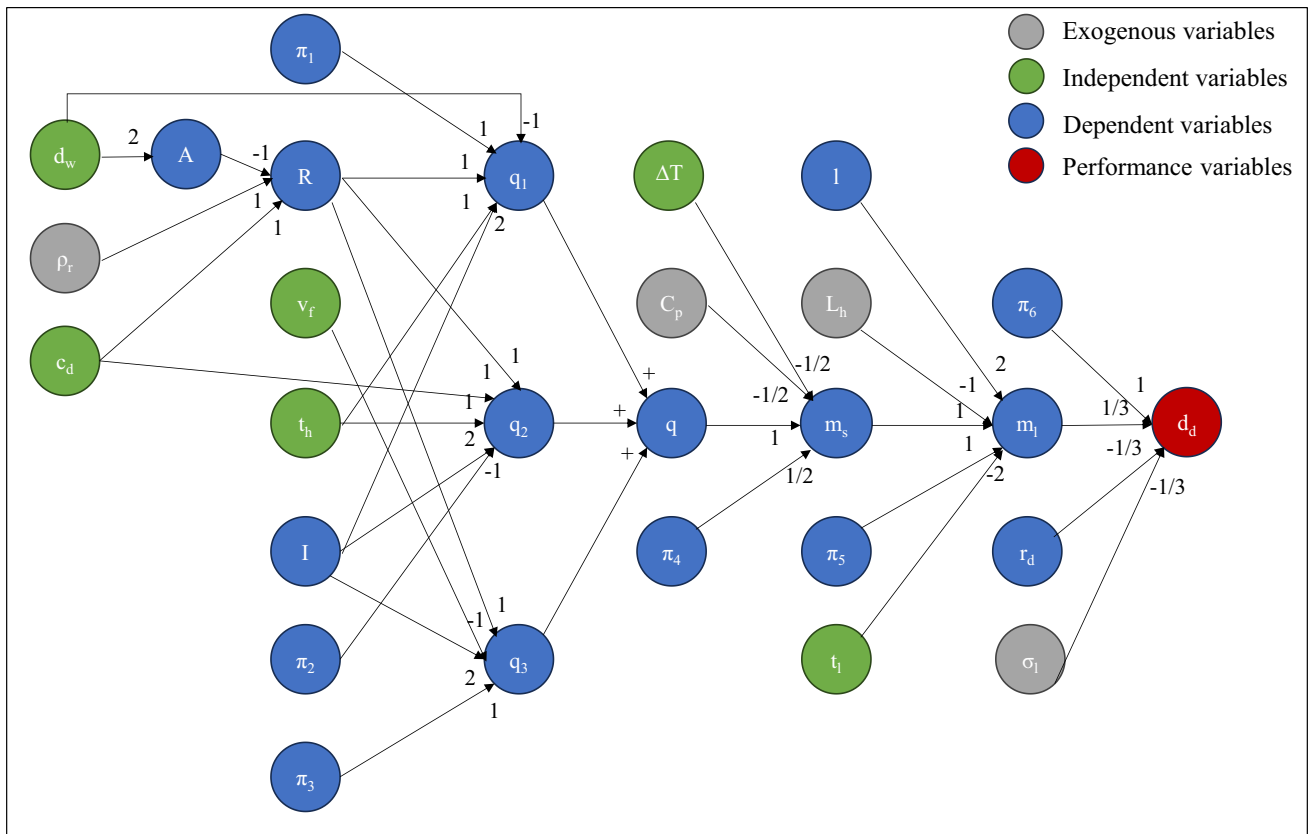


Fig. 7 Causally oriented representation of the GMAW process until the droplet formation

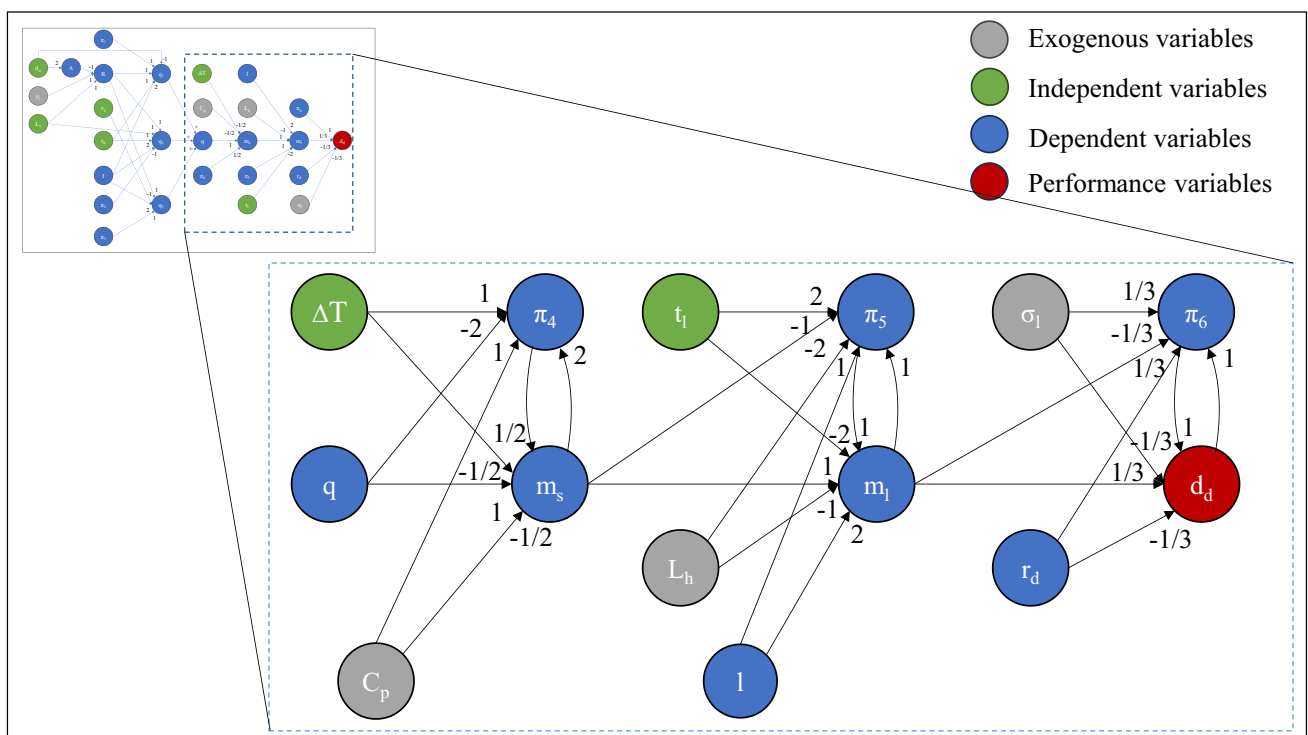


Fig. 8 Section of causally oriented representation with a cyclic relationship between performance variables and respective π numbers

Step 6: Creating system dynamics model in Vensim

Vensim is a platform for creating and simulating system dynamic models. Usually, SD models are built around stocks and flows. The use of SD model proposed in the work is different and exploits directly the causally oriented graph architecture in Vensim. It allows to use the result of step 5 as a direct input to the SD model and to quantitatively analyse and understand droplet formation in GMAW.

The variables of the causally oriented graph are defined using the variable tool in Vensim. Only two types of variables (i.e. excluding stocks and flows) were used in the model: auxiliary and constant variables. Auxiliary variables are computed from other variables at a given time [40]. An auxiliary variable has an expression involving another variable in its equation. Variables such as q , m_s , m_l , and d_d are auxiliary variables represented in the form of expressions containing the values of respective π numbers and the corresponding variables. A constant variable is a variable whose values do not change over time [40]. Constant variables are used to represent parameters, or initial conditions that do not vary during simulation. The input variables, including v_f , I , and r_d along with filler material properties such as R , C_p , T_m , and ρ_l , as well as the d_w , are assigned to constant variables. Along with the variable tool, the simultaneous function is employed in the system dynamics model. Simultaneous function is used where the left-hand side depends, directly or indirectly, on the right-hand side, without any intervening levels [40]. The functions enable a quantitative analysis of the relationship between q , m_s , m_l , and d_d and their respective π numbers. In Vensim, the simultaneous function is represented by cyclic arrows connecting variables and π numbers, which indicate bidirectional influences.

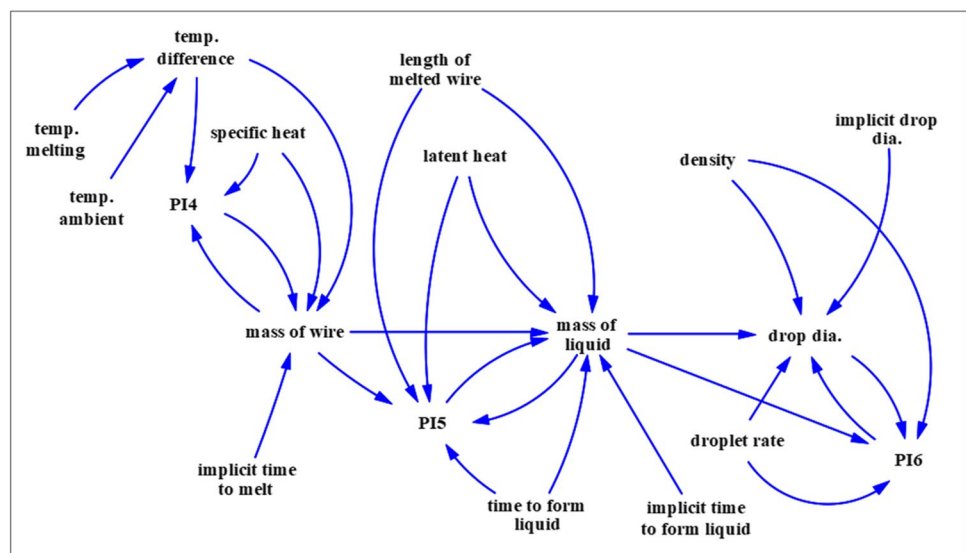
A single arrow signifies the relationship between a constant variable and other variables. The dependencies are indicated by the direction of the arrowhead, which represents the flow of influence from the constant variable to the performance variables. These functions in Vensim are used to create two system dynamic models. The first model illustrated in Fig. 9 contains the simultaneous function between q , m_s , m_l , and d_d and their respective π numbers. This model is used to estimate the unknown π values.

The first model provides an understanding of quantitative change between the values of π numbers and the performance variables. As a result, it could be used for estimating the values of the π numbers.

Fig. 10 shows the second model, used for prediction purpose, q , m_s , m_l , and d_d are the auxiliary variables. The π numbers, input variables, filler material properties, and filler material dimension are considered constant variables. The performance variables are represented as auxiliary variables, each containing an equation specified as a function of their respective π numbers and constant variables. The consecutive auxiliary variables are not just depending on the values of respective π numbers, but also, on the values of prior auxiliary variables. Thus, the interdependence of the auxiliary variables indicates a causally oriented relationship between them. A variation in the input variables would cause a change in the performance variable. As a result, the second model provides an insight on the dependencies that occur during the droplet formation and is appropriate for determining droplet diameter values for a variety of input values. Step 8 of this Section goes into greater detail about the use of the first and second models.

Step 7: Estimation and validating

Fig. 9 Representing cyclic causally orientation graph of Fig. 8 in Vensim



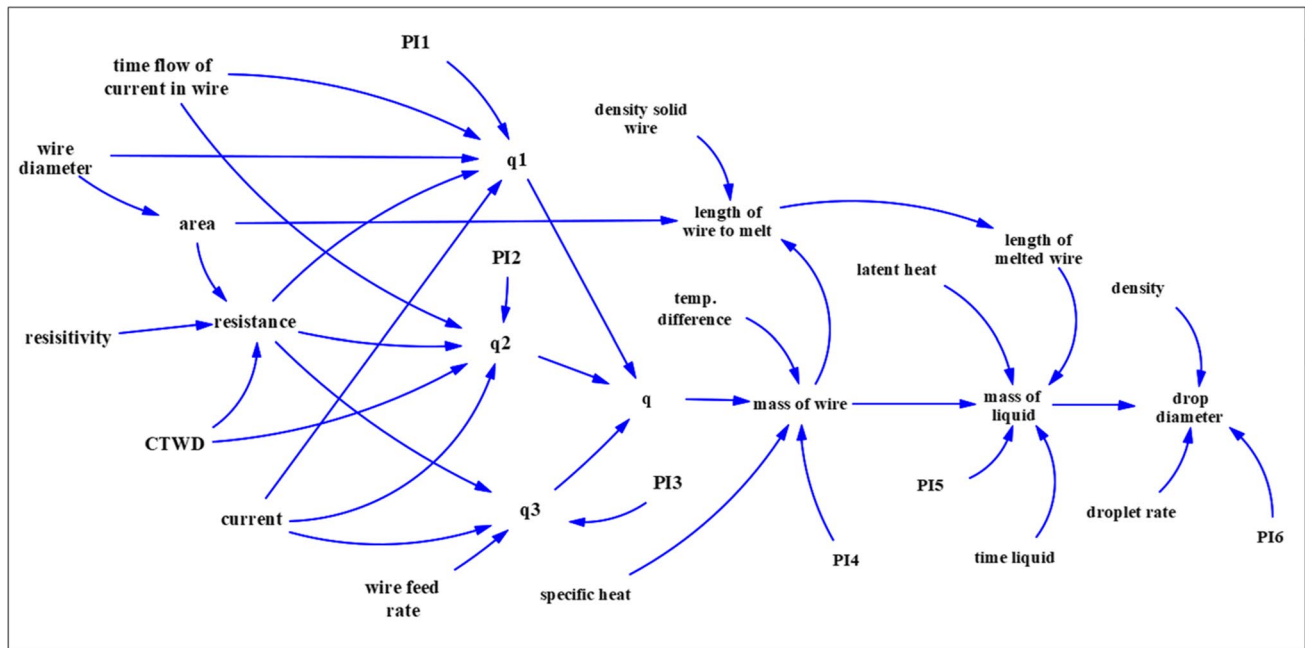


Fig. 10 Representing causally orientation graph of Fig. 7 in Vensim

Table 3 Dataset from article [41]

No	WFR (m/S)	Current (A)	Droplet rate (1/s)	Drop diameter (mm)
1	0.065	178	16	1.29 ± 0.25
2	0.068	180	30	1.08 ± 0.12
3	0.072	194	34	1.05 ± 0.13
4	0.076	206	42	0.99 ± 0.16
5	0.080	211	60	0.90 ± 0.16
6	0.085	216	83	0.82 ± 0.12
7	0.088	225	153	0.68 ± 0.12
8	0.093	229	183	0.65 ± 0.13
9	0.097	232	374	0.52 ± 0.15
10	0.101	246	414	0.51 ± 0.15

Table 4 Filler material properties (ER70S-6)

Properties (units)	Symbols	Values
Surface tension (N m^{-1})	γ	1.67
Density (kg m^{-3})	ρ	7200.00
Density of the solid (kg m^{-3})	ρ_s	7800.00
Viscosity ($\text{kg m}^{-1} \text{s}^{-1}$)	μ	0.006
Specific heat ($\text{J kg}^{-1} \text{K}^{-1}$)	C_p	753.00
Solid specific heat ($\text{J kg}^{-1} \text{K}^{-1}$)	C_{ps}	753.00
Melting temperature (K)	T_m	1798.00
Enthalpy of melting (J kg^{-1})	L_h	2.77E-05
Thermal conductivity ($\text{W m}^{-1} \text{K}^{-1}$)	λ	26.00
Solid thermal conductivity ($\text{W m}^{-1} \text{K}^{-1}$)	λ_s	26.00

The π values are estimated using an experimental dataset. The dataset from [41] was found to be suitable for model estimation and validation. The authors of [41] evaluated the changes in droplet diameter and velocity with wire feed (WFR) and welding current. Their findings indicated that the geometrical shape component of the droplet varies with welding current, and the changes are significant. The dataset in [41] provides ten examples with information on wire feed, welding current, droplet rate, and droplet diameter. This dataset is the unique necessary resource for estimation π values and evaluating models. Readers will note that the size of the dataset is very small and is usually not sufficient for training traditional ANNs (Table 3).

There are variables that represent material properties, process parameters, and filler materials quality. These variables should be configured to match the specification in [41]. The filler material used in [41] is ER70S-6 with a wire diameter of 1.2 mm. The authors refers to Chapuis et al. [13] for data regarding the filler material properties which are reproduced below in Table 4.

In Sect. 2, within Step 7 of the methodology, it has been emphasised that single sample is enough to estimate π values using the model. The problem described in Eqs. (4) to (9). has 12 unknowns and 6 equations. This makes the structure not self-contained. As summarized by Iwasaki and Simon [39], supplementary considerations

are necessary to make it self-contained. Thus, the authors have embedded in their strategy to form the dimensionless numbers, the following principles:

- i. The principle of ideality from TRIZ [42] has been considered together with the law of conservation of energy [43]. This was used also to select the performance variables, form the dimensionless numbers π_1 , π_2 , and π_3 , and set the values of π_i number to 1.
- ii. As an application of the law of conservation, the total mass in a system remains constant. In a well-functioning welding process, the wire feed rate (meter per second) is equivalent to the amount of material melted per second. Thus, the length of wire feed per second equals wire melted per second. Using Eq. (13), we can calculate the assumed value of the mass of solid wire melted (m_s).

$$m_s = \pi \cdot r^2 \cdot h \cdot \rho_s \quad (16)$$

where h = length of wire feed per second, π roughly equal to 3.14 and should not be mixed with the other π_i values in the other equations of this article, ρ_s = density of solid wire, and r = radius of filler wire

- iii. Furthermore, the mass of the solid melted (m_s) is equal to the mass of liquid material (m_l) formed. According to the law of conservation of mass.

$$m_s = m_l \quad (17)$$

- iv. The estimated π_i values are considered constant after estimation and the predictive model uses these constant π_i values. This principle is the integration of the similarity commonly used in scaling principles in DA theory [31].

It should be noted by the readers that this procedure aims to minimize the need for big training dataset. Nevertheless, if the principle of conservation is not considered in the article, the same results can be obtained by using more examples to estimate π values. This will replace the explicit use of the conservation principle and will simply impose to also use simultaneous equations for computing π_1 , π_2 , and π_3 .

The next stage involves determining the values of π_i using the first model from Sect. 3 (Fig. 9). Table 3 contains specific values for variables such as wire feed rate, weld current, and droplet rate for the first example, which are used as input variables for the model, after inputting the input variables, filler material properties, and droplet diameter, into the first model. The value of q is estimated based on the assumed values of π_1 , π_2 , and π_3 . The m_s is determined using the calculated value of q , using principle ii. Consequently, the value of m_s , combined with q , is used to calculate the value of π_4 . Principle iii states that m_s is equal to m_l formed, which allows the calculation of the value of π_5 . To calculate π_6 , the droplet diameter and m_l are used. Using simultaneous function means that the changes in π values will affect the corresponding performance variables, and vice versa. As a result, the values

Table 5 π number values

π no	π_1	π_2	π_3	π_4	π_5	π_6
Value	1	1	1	1.30E-05	0.006	0.51

Table 6 Results

No	WFR (m/S)	Current (A)	Droplet rate (1/s)	π_1	π_2	π_3	π_4	π_5	π_6	Drop diameter Predicted (mm)	Drop diameter from dataset (mm)	Squared error
1	0.065	178	16	1	1	1	1.3E-05	0.006	0.51	1.29	1.29	0.0000
2	0.068	180	30	1	1	1	1.3E-05	0.006	0.51	1.07	1.08	0.0001
3	0.072	194	34	1	1	1	1.3E-05	0.006	0.51	1.10	1.05	0.0025
4	0.076	206	42	1	1	1	1.3E-05	0.006	0.51	1.09	0.99	0.0094
5	0.080	211	60	1	1	1	1.3E-05	0.006	0.51	1.00	0.9	0.0095
6	0.085	216	83	1	1	1	1.3E-05	0.006	0.51	0.93	0.82	0.0116
7	0.088	225	153	1	1	1	1.3E-05	0.006	0.51	0.79	0.68	0.0114
8	0.093	229	183	1	1	1	1.3E-05	0.006	0.51	0.76	0.65	0.0129
9	0.097	232	374	1	1	1	1.3E-05	0.006	0.51	0.62	0.52	0.0091
10	0.101	246	414	1	1	1	1.3E-05	0.006	0.51	0.63	0.51	0.0137
Mean squared error												0.0080

of π numbers are calculated even if the input variables, output variables, and filler material properties change. This is leading to a convergence toward stable values of π_4 , π_5 , and π_6 . Table 5 shows the resulting values of the π numbers.

These π numbers estimated using the model of Fig. 9 are used as inputs to the second model (Fig. 10) to predict the droplet diameter for other examples. The result obtained from these models is shown in Table 6.

From Table 6, it is evident the square errors for examples 2, 3, 4, 5, and 9 are low in comparison with the tolerances associated with the experimental dataset in Table 3. However, with further observation, the examples 6, 7, 8, and 10. They exhibit a moderately high squared error value. This observation could suggest that a change in the behavioural mode of the process might occur at such WFR and current. One way to test the hypothesis is to use other examples from Table 3 to evaluate the impact on squared errors and on the mean squared errors. This is acting as a sensitivity analysis to determine the impact

of the selected examples on the predictions. Table 7 shows the estimated π values for three different examples selected from the dataset corresponding respectively to rows 1, 5, and 8 in Table 3. Table 7 shows the variations amongst the estimated values of π_4 , π_5 , and π_6 . It is interesting to note that π_4 and π_5 , respectively, related to the function “to melt wire” in Fig. 5 are significantly affected by the change in the input values. This might indicate that the melting process is going through variations due to the increase of WFR and current. This should indicate that the similarity principle *iv* considered is a rather rough initial approximation. This sensitivity analysis could be used in future work to cluster the experimental dataset into different behavioural modes.

Furthermore, the sensitivity analysis provides judgment for different phases of model development [44]. Thus, it is used in this article as a technique for model validation. Table 8 shows a remarkably robust prediction of the droplet diameter for the selected examples. The mean square errors are low for the three selected example. Thus, the

Table 7 π numbers value for the three different examples

	First example	Fifth example	Eighth example	Standard deviation	Range of the variation compared with the mean values
π no	Value	Value	Value		
π_1	1	1	1	0	0
π_2	1	1	1	0	0
π_3	1	1	1	0	0
π_4	1.3E-05	1.9E-05	2.5E-05	8.5E-06	44,7%
π_5	0.006	0.0043	0.0032	0.0014	31,4%
π_6	0.51	0.48	0.48	0.017	3.5%

Table 8 Prediction result for the three different examples

No	First example			Fifth example			Eighth example		
	Drop diameter predicted (mm)	Drop diameter from dataset (mm)	Squared error	Drop diameter predicted (mm)	Drop diameter from dataset (mm)	Squared error	Drop diameter predicted (mm)	Drop diameter from dataset (mm)	Squared error
1	1.29	1.29	0.0000	1.16	1.29	0.0166	1.11	1.29	0.0335
2	1.07	1.08	0.0001	0.96	1.08	0.0137	0.92	1.08	0.0262
3	1.10	1.05	0.0025	0.99	1.05	0.0037	0.94	1.05	0.0113
4	1.09	0.99	0.0094	0.98	0.99	0.0002	0.93	0.99	0.0034
5	1.00	0.9	0.0095	0.90	0.9	0.0000	0.86	0.9	0.0020
6	0.93	0.82	0.0116	0.83	0.82	0.0002	0.80	0.82	0.0006
7	0.79	0.68	0.0114	0.71	0.68	0.0008	0.67	0.68	0.0000
8	0.76	0.65	0.0129	0.69	0.65	0.0014	0.65	0.65	0.0000
9	0.62	0.52	0.0091	0.55	0.52	0.0011	0.53	0.52	0.0001
10	0.63	0.51	0.0137	0.56	0.51	0.0029	0.54	0.51	0.0008
	Mean square error		0.008	Mean square error		0.004	Mean square error		0.0077

model has a high sensitivity, to the change in the input parameter. This would be a desirable attribute when the model is used for online control of the GMAW process. The π numbers behave as biases if an analogy with ANNs is considered. This could mean the prediction quality depends on the values of π numbers. The best predictive result is obtained for example five corresponding probably to experimental values at the limit of a specific functioning mode. It tends to show that also some rules and principles can be extracted from this study to select the most appropriate example limiting the prediction means square error.

4 Discussion

The approach described in this paper has been developed as an extension of the DACM approach to cope with quantitative predictions in a context where experimental datasets are expensive and difficult to generate. Such an approach could potentially bring a significant economic benefit and have significant environmental implications. Indeed, if future research will be conclusive, it can be a way to generate parsimonious predictive models, which are explainable and can be trained with a limited computing power and with very small number of samples.

The long-term goal of this research is to produce models that can be incorporated in the control systems of manufacturing equipment for real-time control of the processes. For this reason, the response time of the control system is an important factor to consider, and the approach presented in this paper; we believe can be easily deployed together with edge computing principles that will facilitate real-time control by improving response time and save bandwidth of the manufacturing systems.

Unlike most ML algorithms employed currently that need significant datasets to be trained, the presented approach exploiting and integrating different supplementary sources of knowledge significantly reduce the need for big datasets. The variations in values especially for π_4 and π_5 in Table 7 reveal a probable variation in the process physics taking place after experiment five in Table 3.

The approach presented in the paper facilitates the analysis of the mass transfer phenomena taking place in the welding. It is notable that the function to melt wire is most affected by the sensitivity analysis made in this study. This is giving an indication that exploring the physics associated with this function might be a supplementary study to be conducted. A well-known system engineering principle is the change of viewpoint. In future studies a change from functional viewpoint to physics-based viewpoint for this specific function can help the understanding and analysis. The focus will be the thermodynamic aspects related to the

melting process. One idea is to build new dimensionless numbers associated with the specific variables considered in π_4 and π_5 but specifically focusing on the description of the physics involved with this specific function of the process.

The concept of capability index such as process capability index (C_{pk}) or production performance index (P_{pk}) is used extensively in statistical process control (SPC) methods to support the selection of production equipment capable of producing parts with specified requirements [45]. A novelty proposed in this article is to transfer the concepts of P_{pk} and C_{pk} to evaluate the performance of the predictive model produced in the article. P_{pk} is used when the process is in control and is stable [45]. As, it is not possible to establish that the process is within control and is stable. The C_{pk} indicator is appropriate for our application.

The nature of the statistical distribution is not considered in the study due to the limited size of the experimental sample. But the central limit theorem is applicable in this context, a coherent hypothesis is to consider a normal distribution. The C_{pk} considers the centering of the process. The C_{pk} is computed using (18) where the upper limit (UL), lower limit (LL), mean ($\bar{X}$), and the standard deviation (σ) of the process. A process must have a $C_{pk} > 1$ to be considered a capable process.

$$C_{pk} = \min\left(\frac{UL - \bar{X}}{3 \times \sigma}, \frac{\bar{X} - LL}{3 \times \sigma}\right) \quad (18)$$

Shown in Fig. 11, upper limit (UL), and lower limit (LL) of the dataset from Table 3, where the points 1.29 and point 0.53 represent the highest and lowest predicted values, respectively, with $\bar{X}$ representing the mean of the predicted values from Table 8, where $UL = 1.29 + 0.25 = 1.54$ and $LL = 0.51 - 0.15 = 0.36$. Based on this information, the following observations can be drawn. Firstly, the upper and lower limits of the predicted values lie inside the upper and the lower limits values obtained from dataset. This suggests that the values of the predict droplet diameter are clearly within the limits of the dataset. Secondly, the means of the predicted values appear to be centred i.e. between the upper and lower limits of the dataset. Therefore, from these

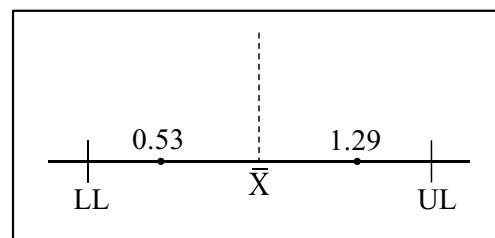


Fig. 11 Number line representation of the prediction model

Table 9 Clusters based on relative error might be representing change in behaviours

No.	First example			Fifth example			Eighth example		
	Drop Diameter Predicted (mm)	Drop Diameter from dataset (mm)	Relative error	Drop Diameter Predicted (mm)	Drop Diameter from dataset (mm)	Relative error	Drop Diameter Predicted (mm)	Drop Diameter from dataset (mm)	Relative error
1	1,29	1,29	0 %	1,16	1,29	10 %	1,11	1,29	14 %
2	1,07	1,08	1 %	0,96	1,08	11 %	0,92	1,08	15 %
3	1,10	1,05	5 %	0,99	1,05	6 %	0,94	1,05	10 %
4	1,09	0,99	10 %	0,98	0,99	1 %	0,93	0,99	6 %
5	1,00	0,9	11 %	0,90	0,9	0 %	0,86	0,9	5 %
6	0,93	0,82	13 %	0,83	0,82	2 %	0,80	0,82	3 %
7	0,79	0,68	16 %	0,71	0,68	4 %	0,67	0,68	1 %
8	0,76	0,65	17 %	0,69	0,65	6 %	0,65	0,65	0 %
9	0,62	0,52	18 %	0,55	0,52	6 %	0,53	0,52	2 %
10	0,63	0,51	23 %	0,56	0,51	11 %	0,54	0,51	5 %

observations, we can draw a conclusion that C_{pk} for our model has to be greater than 1.

Another idea emerging from this study is also to use the sensitivity analysis as a clustering principle for the dataset. A possible analytical approach to detect different functioning modes is to start clustering the experimental dataset by detecting the variations of the π number's values. The relative error in prediction could be used as a metric to identify clusters with a possible limit in relative error that can be determined to highlight clusters. In Table 9, a cluster should have a relative error lower or equal to 5%. Thus, forming possible clusters highlighted in light green in Table 9. It is noticeable that the choice of the selected examples affects the scope of the model. There could be potentially two to three types of distinct behaviours. It is probable that a change is taking place between globular to spray modes with a stage where the transfer is transitioning between those two transfer modes.

Discovering the behavioural modes in GMAW can help selecting the most appropriate example to generate robust and accurate models at the end of the process. This can drive the creation of several predictive models and to the possibility from switching from one model to the other depending on the input values.

The accuracy of the model produced is very good and the model has demonstrated a rather robust behaviour. This is due to the fact that multiple layers of knowledge have been integrated in the model as a result of the process proposed in the paper.

The next step of the work is also to integrate the study of the melt pool in the current model as visible in Fig. 5. The authors have indicators showing that the same approach

applied to the melt pool evaluation can also be successful. Indeed, dimension of weld pool could be accurately predicted from the deposited volume of the weld metal [10]. This should be connected with the integration of information coming directly from sensors such as high-speed cameras. Chapuis et al. [13] used high-speed imaging of pulsed gas metal arc welding (P-GMAW) to show the macro drop (weld pool) evolution is driven by heat and mass transfer by the arc plasma and the deposited droplet.

The impact of droplet diameter and droplet frequency, as well as the property of the melt pool, must be correlated with the final weld properties both at geometric and at material properties levels. According to Fan et al. [39], the frequency of droplet transfer and the location at which the droplet impinges on the weld are critical parameters influencing the weld bead's depth and width.

The model developed is useful for predicting droplet diameters in the GMAW process. The model could be used to estimate the quantity of material deposited. However, it cannot determine the velocity of impingement and the temperature of the droplets. Thus, in order to predict the width and depth of the weld pool, future models must include this information.

Despite the correct model predictions, there exist certain limitations that will drive further studies:

- The change in π numbers can be associated with a specific characteristic of the process or physics as presented above. Currently, the model does not integrate a full exploitation of the ideas presented in this paper to detect and model the different behavioural modes.

- ii. The model has been developed with ER70S-6 material and the GMAW process. The model's ability to predict different welding processes and materials has not been tested and this is a major effort and goal to make such type of model as generic as possible. However, the small number of examples required, as well as the generic structure of the model, should be supporting easy and fast transfer of this model to other welding technologies.

5 Conclusion

In summary, the proposed approach combines the DACM framework with system dynamics. DACM is used to integrate and formulate the process knowledge, and create the π numbers and the causally oriented graphs representing the processes. The causally oriented graph forms the foundation for the system dynamics model. The implementation of the system dynamic model in Vensim facilitated the prediction of droplet diameter in GMAW process. It should be noted that several other simulation methods could be employed for simulation purpose too.

The ability of the model to estimate π values with a single example distinguishes it from conventional machine learning algorithms. The proposed modelling methodology is advantageous because of its explainability, parsimony, and its minimal requirement for data. This can have major positive impact both from economic and environmental standpoints. The authors conducted a sensitivity analysis to test the model's fidelity. Three examples were selected. The prediction results demonstrate satisfactory performance with a low mean square error. It gave also additional insight to different behavioural modes (droplet transfer modes) of the GMAW process. It also provided ideas on how to cluster the experimental dataset into separate behavioural clusters. Those insights will be investigated further in future research.

The future studies will involve:

- i. Expanding the scope of the model to predict the height and the width of a single bead in GMAW.
- ii. Including gas flow, magnetic permeability, drag force, and other such welding related variables in the model. That were excluded from the current model.
- iii. Incorporate the dynamic variation in the π numbers and relate these changes to certain behavioural modes and functions of the process.
- iv. Testing the generality and the ease of transfer of the current model on a broad range of welding scenarios and welding technologies involving different process parameters, filler materials, and geometries.
- v. Forming possible hybrid models integrating the current model with other more conventional machine

learning strategies to adapt the current approach to different processes and welding methods including laser welding for example.

Author contribution Akshay Dhalpe: conceptualization, methodology, investigation, formal analysis, writing—original draft, review and editing. Hossein Mokhtarian: methodology, formal analysis, writing—review and editing. Suraj Panicker: investigation, formal analysis, writing—review and editing. Di Wu: investigation, writing—review and editing. Joe David: formal analysis, writing—review and editing. Shahriar Bakrani Balani: formal analysis, writing—review and editing. Jose Martinez Lastra: writing—review and editing, supervision. Eric Coatanéa: conceptualization, methodology, investigation, writing—review and editing, supervision, project administration, funding acquisition.

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Declarations

Competing interests The authors declare no competing interests.

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